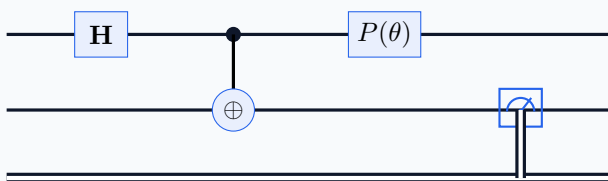


Classical and Quantum Theory for QPU

From Bits to Bloch Spheres



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September 13, 2026

A ground-up explanation of classical logic, complex amplitudes, quantum state, measurement, entanglement, mixed states and noise, and the mathematics represented by QPU gates.

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1 Classical computation foundations

Quantum circuits build on ideas from classical digital logic. Understanding bits, Boolean functions, truth tables, and reversible computation makes the QPU gate model much easier to use.

1.1 Classical bits and logical state

A classical bit has one definite logical value:

$$b \in \{0, 1\}.$$

Physical computers may represent that value with voltage, charge, magnetic orientation, or another measurable quantity. Digital logic ignores most physical detail and works with the abstraction 0 or 1.

A wire carries a bit from one operation to another. A *gate* computes a function of one or more incoming bits. For example, NOT has one input and one output:

$$\text{NOT}(0) = 1, \quad \text{NOT}(1) = 0.$$

In QPU source, a token names the logical wire. On computational-basis states, $X \text{ } -I \text{ } Q \text{ } -0 \text{ } Q$ has the same measured input/output relationship as classical NOT.

1.2 Boolean algebra

Boolean algebra describes functions whose values are 0 and 1. This guide uses:

Operation	Symbol	Spoken form	Meaning
NOT	$\neg A$	not A	1 when A is 0; 0 when A is 1.
AND	$A \wedge B$	A and B	1 only when both inputs are 1.
OR	$A \vee B$	A or B	1 when either or both inputs are 1.
XOR	$A \oplus B$	A exclusive-or B	1 when exactly one input is 1.
NAND	$\neg(A \wedge B)$	not-and	The complement of AND.

For bits, XOR is addition modulo two:

$$0 \oplus 0 = 0, \quad 0 \oplus 1 = 1, \quad 1 \oplus 0 = 1, \quad 1 \oplus 1 = 0.$$

There is no carry in modulo-two addition. Ordinary binary addition handles the carry separately, which is why a half-adder produces both Sum and Carry.

1.3 Truth tables as complete finite specifications

A truth table lists a function's output for every possible input pattern. With n binary inputs there are 2^n patterns. AND, OR, XOR, and NAND can be compared in one table:

A	B	$A \wedge B$	$A \vee B$	$A \oplus B$	$\neg(A \wedge B)$
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	1
1	1	1	1	0	0

For a deterministic combinational circuit, this table completely specifies the classical input/output behavior. It does not reveal the circuit's internal gate arrangement, propagation time, power consumption, or quantum phase.

1.4 Combinational logic and stored state

A *combinational* output depends only on current inputs. Adders and the Boolean gates above are combinational. A *sequential* circuit also depends on stored state from an earlier step, as in counters and latches.

QPU cycle suffixes can document successive stages, and INCREASECYCLE marks a workspace boundary, but the compiled circuit is still one ordered gate sequence. `SAVE_STATE` and `LOAD_STATE` copy and restore the simulator state vector. They are not quantum gates. When modeling stateful behavior inside the unitary circuit, expose previous state as `PARAMS` and return next state through `RETURNVALS`.

1.5 Why ordinary classical gates can lose information

A function is reversible only if each output identifies exactly one input. Classical NOT is reversible: seeing output 1 proves the input was 0. AND is not: output 0 could have come from 00, 01, or 10.

$$(A, B) \xrightarrow{\text{AND}} A \wedge B$$

maps four input patterns onto only two output values, discarding information. Quantum evolution through closed-system gates must be unitary and therefore reversible. A quantum/reversible AND preserves both inputs and mutates another target:

$$(A, B, t) \mapsto (A, B, t \oplus (A \wedge B)).$$

Knowing the output triple lets the transformation be run backward to recover the input triple. This one-to-one property is one of four conditions a quantum gate must satisfy to be reversible; *Requirements for a reversible gate*, in the Theory Guide’s mathematical foundations chapter, states all four together.

How to read $t' = t \oplus (A \wedge B)$

- A and B are **controls**. They are only read. After the gate they still hold exactly what they held before, which is why the input can always be recovered.
- t is the **target**, the output workspace. It is the only wire that changes. The prime in t' means “ t after the gate”.
- $\oplus$ means “flip if”. The target flips when $A \wedge B$ is 1 and stays put otherwise.
- Start the target at 0 and it ends holding $A \wedge B$, the ordinary AND answer. Start it at 1 and it ends holding NAND instead.

In the Circuit builder, A is **Control A**, B is **Control B**, and t is **Target particle**.

1.6 Controls, targets, and workspace

These terms connect classical logic to QPU syntax:

Control A wire whose value decides whether a target operation occurs. In `CNOT -I A -O T`, A is the control.

Target The wire changed by an operation. T is the target in that `CNOT`.

Workspace or ancilla An extra wire introduced to hold an intermediate or output while preserving inputs. It is commonly initialized to $|0\rangle$.

Garbage Intermediate information left in workspace after the desired answer has been computed. Reversible algorithms often *uncompute* temporary gates to return garbage wires to zero.

Uncomputation Applying inverse operations in reverse order to clean workspace without erasing information irreversibly.

The QPU language calls logical wire names tokens. `CREATETOKEN` reserves workspace, `SET` prepares it, `-I` lists inputs/controls, and `-O` names targets.

1.7 Fan-out, copying, and the no-cloning boundary

Classical logic freely copies a bit onto several wires. If the source is a known basis bit and targets begin at zero, `CNOT` reproduces that basis value:

$$|x\rangle|0\rangle \xrightarrow{\text{CNOT}} |x\rangle|x\rangle, \quad x \in \{0, 1\}.$$

For a superposition $\alpha|0\rangle + \beta|1\rangle$, the same circuit produces

$$\alpha|00\rangle + \beta|11\rangle,$$

not

$$(\alpha|0\rangle + \beta|1\rangle) \otimes (\alpha|0\rangle + \beta|1\rangle).$$

The first state is generally entangled; the second would be two independent copies and contains cross terms. The no-cloning theorem forbids a universal gate that copies every unknown quantum state. QPU aliases similarly give two names to one wire; they do not create a physical copy.

1.8 Deterministic, probabilistic, and quantum models

Model	State description	Transformation	Observation
Deterministic classical	One definite bit string	Boolean function	The definite output bits
Probabilistic classical	Probabilities over bit strings	Stochastic update	A sampled string
Pure-state quantum	Complex amplitudes over basis strings	Unitary matrix	A sampled string after magnitude-squared probabilities and collapse

Negative and imaginary amplitudes are not negative or imaginary probabilities. They carry phase. Two computational paths can add constructively or cancel destructively before probabilities are calculated. This interference has no counterpart in an ordinary classical random-choice table.

1.9 Classical and QPU operation map

Classical idea	QPU operation	Important difference
NOT	X or NOT	Acts linearly on superpositions; X is unitary.
XOR into target	CNOT	Preserves the control and can entangle.
AND into target	CCNOT or AND	Preserves controls; target is XOR-mutated.
Swap variables	SWAP	Exchanges complete quantum wire states.
Random bit	H then MEASURE	H first creates coherent amplitudes, not an ordinary hidden random choice.
Read a bit	MEASURE	Sampling collapses a superposition and can disturb entanglement.
Initialize zero	SET token 0p	Compiles to state preparation/reset behavior.
Function output	RETURNVALS	Declares an interface; does not itself copy or measure a wire.

2 Mathematical foundations

This chapter assumes no prior quantum mechanics. The essential ideas are numbers, vectors, probability, and transformations. Quantum notation can look unfamiliar, but each symbol has a concrete computational meaning.

2.1 Real numbers and complex numbers

The real numbers include familiar values such as -2 , 0 , $1/3$, and $\sqrt{2}$. A real number can be placed on a one-dimensional number line. Quantum amplitudes require a second numerical direction, represented by the *imaginary unit*

$$i = \sqrt{-1}, \quad i^2 = -1.$$

There is no real number whose square is -1 , so i extends the number system. A complex number has the form

$$z = a + bi,$$

where a is the real part and b is the imaginary part. Examples are $3 + 2i$, $-i$, and 1 (which is $1 + 0i$).

Complex arithmetic follows ordinary algebra together with $i^2 = -1$:

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

For example,

$$(1 + i)^2 = 1 + 2i + i^2 = 2i.$$

2.2 The complex plane, magnitude, and conjugate

A complex number is a point or arrow in a plane: horizontal coordinate a , vertical coordinate b . Its magnitude is the arrow length,

$$|z| = \sqrt{a^2 + b^2}.$$

The complex conjugate reflects the arrow across the real axis:

$$z^* = a - bi.$$

Multiplying by the conjugate gives a nonnegative real number:

$$z^*z = (a - bi)(a + bi) = a^2 + b^2 = |z|^2.$$

Quantum measurement probabilities use $|z|^2$, not z itself. This guarantees real, nonnegative probabilities even though amplitudes may be complex.

2.3 Angles, radians, sine, and cosine

An arrow in the complex plane may be described by its length r and angle θ :

$$a = r \cos \theta, \quad b = r \sin \theta.$$

Quantum formulas normally measure angles in radians. One full turn is 2π radians, a half turn is π , and a quarter turn is $\pi/2$. Conversion is

$$\theta_{\text{radians}} = \theta_{\text{degrees}} \frac{\pi}{180}.$$

This is why `PHASE=90d` and `PHASE=pi/2` describe the same angle.

2.4 Euler's formula and Euler's identity

Euler's formula connects complex numbers to rotations:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

The value $e^{i\theta}$ always has magnitude one. As θ changes, it moves around the unit circle in the complex plane. Important cases are

$$e^{i0} = 1, \quad e^{i\pi/2} = i, \quad e^{i\pi} = -1, \quad e^{i2\pi} = 1.$$

Setting $\theta = \pi$ gives Euler's identity,

$$e^{i\pi} + 1 = 0.$$

In a phase gate, multiplying an amplitude by $e^{i\theta}$ rotates that amplitude without changing its magnitude. Its immediate measurement probability is unchanged, but its relationship to other amplitudes changes. Later gates can convert that relative phase into different probabilities.

2.5 Vectors and basis vectors

A column vector is an ordered list of numbers. The two basis vectors used for one qubit are

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The vertical bars and angle bracket form Dirac's *ket* notation. `0p` prepares $|0\rangle$ and `1p` prepares $|1\rangle$. Any one-qubit pure state is a complex linear combination:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

The complex numbers α and β are *probability amplitudes*. They are not probabilities themselves.

2.6 Normalization and the Born rule

A valid state is normalized:

$$|\alpha|^2 + |\beta|^2 = 1.$$

The Born rule turns amplitudes into measurement probabilities:

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2.$$

For the plus state,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

both amplitudes are $1/\sqrt{2}$. Squaring their magnitudes gives $P(0) = P(1) = 1/2$. The factor $1/\sqrt{2}$ is therefore required; writing $|0\rangle + |1\rangle$ without normalization would produce total probability two.

2.7 Matrices as gates

A matrix is a rectangular array of numbers. A one-qubit gate is represented by a 2×2 complex matrix. Matrix-vector multiplication computes the new amplitudes:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{bmatrix}.$$

For example, the X gate exchanges the two amplitudes:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}.$$

Applying gates in source order means multiplying by matrices in the same temporal order: if U acts first and V second, then

$$|\psi_{\text{final}}\rangle = V(U|\psi_{\text{initial}}\rangle) = VU|\psi_{\text{initial}}\rangle.$$

2.8 Requirements for a reversible gate

A quantum gate is, by default, reversible: closed-system evolution is unitary, and unitarity already implies everything “reversible” requires. The four conditions below are one fact, stated four ways — as a matrix equation, as a statement about information, as a wire count, and as a scope limit — so that each can be checked on its own.

1. **Unitarity.** The gate's matrix U must satisfy

$$U^\dagger U = U U^\dagger = I,$$

where $U^\dagger$ transposes U and complex-conjugates every entry (*The complex plane, magnitude, and conjugate*), and I is the identity matrix. Unitarity guarantees an inverse, $U^{-1} = U^\dagger$: the gate's action can always be undone by running $U^\dagger$. It is also exactly the condition that keeps a state normalized ($|\alpha|^2 + |\beta|^2 = 1$, *Normalization and the Born rule*) after the gate runs, so the total probability of all outcomes stays 100%.

2. **Bijective mapping.** The gate must be a one-to-one map from input states to output states: given only the output and the gate, the input can always be reconstructed. Classical NOT has this property (an output of 1 proves the input was 0); classical AND and OR do not (an output of 0 does not say which inputs produced it), which is why they are not themselves reversible. A gate satisfying $U^\dagger U = I$ automatically has this property, because $U^\dagger$ is precisely the map that reconstructs the input from the output. Unitarity and bijectivity are the same fact viewed two ways; a reversible gate never erases information.
3. **Equal inputs and outputs.** U is a square matrix, so a reversible gate has exactly as many output wires as input wires: no wire can be discarded mid-computation without breaking the bijection above. This is why QPU's AND is not the two-input classical gate but a three-wire embedding,

$$|a, b, t\rangle \mapsto |a, b, t \oplus (a \wedge b)\rangle,$$

the doubly-controlled CCNOT/Toffoli gate (Doubly controlled NOT / Toffoli). With t starting at $|0\rangle$, $t' = a \wedge b$: the classical AND is recovered on t' without erasing a or b , at the cost of one extra wire the two-input classical gate did not need.

4. **Isolation from measurement.** The three conditions above hold only for a closed system, with no interaction with the environment and, in particular, no MEASURE (Measurement gate). Measurement is not linear and has no inverse matrix: it collapses a superposition to one classical outcome and permanently discards the amplitudes that were not observed. RESET shares this — preparing a wire at $|0\rangle$ regardless of its prior state is not invertible either (Reversible gates versus RESET). MEASURE and RESET are legitimate operations, just not reversible ones, and they are the only two exceptions to an otherwise-unitary gate list.

3 Qubits, phase, and the Bloch sphere

3.1 A qubit is not just an unknown bit

A classical bit is exactly 0 or 1. A qubit may be in any normalized superposition $\alpha|0\rangle + \beta|1\rangle$. This does not mean that the qubit secretly chose a classical value before measurement. Relative amplitudes can interfere, producing effects no ordinary random bit can reproduce.

3.2 Global phase and relative phase

Multiplying an entire state by a unit complex number does not change physical predictions:

$$|\psi\rangle \quad \text{and} \quad e^{i\gamma}|\psi\rangle$$

have the same measurement probabilities and are considered equivalent up to a *global phase*. Relative phase is different. The states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

give equal 0/1 probabilities if measured immediately, yet H sends $|+\rangle \mapsto |0\rangle$ and $|-\rangle \mapsto |1\rangle$. Their relative sign is observable through interference.

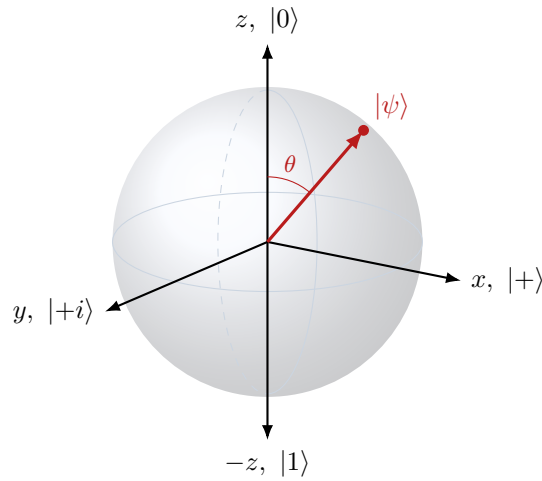
3.3 Bloch-sphere coordinates

After removing an unobservable global phase, every pure one-qubit state can be written with two real angles:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle,$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$. These angles define a point on the unit Bloch sphere:

$$x = \sin \theta \cos \phi, \quad y = \sin \theta \sin \phi, \quad z = \cos \theta.$$



The north pole is $|0\rangle$ and the south pole is $|1\rangle$. The equator contains equal-magnitude superpositions. $|+\rangle$ points along $+x$, $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ along $-x$, and $|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$ along $+y$.

3.4 How gates move a Bloch vector

Ignoring global phase, many one-qubit gates are rotations:

- X rotates the sphere by π around the x-axis, exchanging north and south.
- Y rotates by π around the y-axis.
- Z rotates by π around the z-axis, changing relative phase.
- S and T rotate around z by $\pi/2$ and $\pi/4$.
- PHASE rotates around z by its supplied angle θ .
- H rotates the coordinate description so the z basis and x basis are exchanged: $|0\rangle \leftrightarrow |+\rangle$ and $|1\rangle \leftrightarrow |-\rangle$.

The sphere is a teaching model for one pure qubit. It cannot display the full state of entangled multi-qubit systems as one ordinary three-dimensional arrow. Each particle card instead draws its wire's own reduced state: the coordinates are the measurable averages $x = \langle X \rangle$, $y = \langle Y \rangle$, $z = \langle Z \rangle$, and a wire entangled with others points to the inside of the sphere ($r < 1$). The chapter Mixed states, entanglement measures, and noise explains why.

4 Multiple qubits and entanglement

4.1 Tensor products

Two classical bits have four possibilities; two qubits have four basis vectors:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

Independent systems combine with the tensor product $\otimes$. If $|A\rangle = [a_0, a_1]^T$ and $|B\rangle = [b_0, b_1]^T$, then

$$|A\rangle \otimes |B\rangle = \begin{bmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{bmatrix}.$$

For n qubits, the simulator stores 2^n amplitudes:

$$|\Psi\rangle = \sum_{x=0}^{2^n-1} a_x |x\rangle, \quad \sum_x |a_x|^2 = 1.$$

This exponential growth is why state-vector simulation becomes expensive as qubit count increases.

4.2 Entanglement

Some multi-qubit states cannot be factored into independent one-qubit states. The Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

is entangled. Each qubit measured alone is equally likely to be 0 or 1, but their outcomes are perfectly correlated. It can be prepared from $|00\rangle$:

```
H -I A -0 A
CNOT -I A -0 B
```

The first gate creates a superposition; the controlled gate correlates B with A. Entanglement is not a classical alias. It is a property of the joint amplitude vector.

Correlated is not the same as copied

After the *Bell state* starter circuit, both wires always measure the same value, 00 or 11, each half the time. That looks like copying, but it is not. Before measurement neither wire has a value of its own: stop just before the first meter with **Step gate**, and the **Final state vector** lists only basis states in which q0 and q1 agree, with no separate state for either wire. Choose **Reset state** and **Run all** several times and the pair changes together between 00 and 11.

4.3 Controlled operations

A controlled gate applies a target operation only on basis components where its control condition is true. CNOT applies X when its control is one. If the control is in superposition, linearity applies the operation to one branch and not the other, which can create entanglement:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

CCNOT requires both controls to be one. The system's OR and XOR derived gates use “any active” and “odd parity” predicates respectively.

5 Measurement, probability, and truth tables

5.1 Projective measurement

For a measured qubit q in a multi-qubit state,

$$P(q = 1) = \sum_{x: x_q=1} |a_x|^2, \quad P(q = 0) = 1 - P(q = 1).$$

The simulator samples one outcome. Amplitudes incompatible with that outcome become zero; compatible amplitudes are divided by the square root of the kept probability so the state is normalized again. Measurement is destructive in the sense that the pre-measurement superposition is generally unavailable after collapse.

5.2 Single runs and repeated runs

One simulation run produces one measurement sample. To estimate a probability, run the same preparation and circuit many times. If a plus state is measured 1 approximately 500 times in 1000 shots, the observed frequency 500/1000 estimates the theoretical probability 1/2. More shots reduce ordinary sampling noise but never reveal a complex amplitude directly.

5.3 Truth tables versus quantum amplitudes

A classical truth table specifies output bits for listed input bits. It is a good contract for arithmetic and reversible Boolean circuits operating on computational-basis states. It cannot fully describe phase, interference, entanglement, or a probability distribution.

The QPU truth-table runner prepares each input row, executes the circuit, measures declared RETURNVALS, and compares those measured bits. In companion .qpuio tables, cells are 0p, 1p, or sp. On an expected output, sp means “either measured bit is acceptable.” Use state equations and repeated experiments in addition to tables when a circuit’s purpose is genuinely quantum.

5.4 A complete interference example

Starting in zero, two Hadamards restore zero:

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{H} |0\rangle.$$

Inserting Z changes the relative phase and changes the final answer:

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z} |-\rangle \xrightarrow{H} |1\rangle.$$

```
MAIN-PROCESS Interference
CREATETOKEN -I Q
SET Q 0p
H -I Q -0 Q
Z -I Q -0 Q
H -I Q -0 Q
MEASURE -I Q
RETURNVALS Q
```

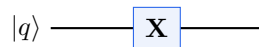
The middle Z does not change an immediate 0/1 probability, yet interference at the final H makes its effect deterministic. This is the central reason the system documents matrices and phase as well as truth tables.

6 Single-qubit primitive gates

Placing a single-qubit gate

X, Y, Z, H, S, T, RX, RY, RZ, and PHASE act on one wire, so the workbench uses only **Target particle**; **Control A** and **Control B** are ignored. PHASE, RX, RY, RZ, and CPHASE also show a **Phase angle** slider. On the canvas, drop the block on the wire you want to change.

6.1 Pauli X (X)



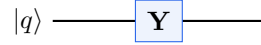
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \quad X(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle.$$

X is the quantum bit flip. On the Bloch sphere it is a half-turn around the x-axis. It exchanges the north and south poles and is self-inverse: $X^2 = I$. Unlike a classical NOT implementation, X remains meaningful on a superposition because it exchanges amplitudes linearly without measuring.

```
X -I Q:0 -0 Q:0
```

Input	Classical comparison	Output
0	NOT 0	1
1	NOT 1	0

6.2 Pauli Y (Y)



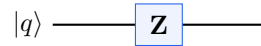
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle.$$

It flips the basis value like X but adds phases that become observable through later interference. Geometrically it is a half-turn around the y-axis and is self-inverse, $Y^2 = I$. The factors i and $-i$ are global phases for either basis state by itself, but on components of a larger superposition they may become relative phases.

Y -I Q -0 Q

Input	Measured bit	Quantum output
0	1	$i 1\rangle$
1	0	$-i 0\rangle$

6.3 Pauli Z (Z)



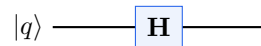
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

Z does not change a computational-basis measurement, but it changes relative phase. For example, $Z|+\rangle = |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. On the Bloch sphere it is a half-turn around z. It is self-inverse and is the simplest example showing why quantum state cannot be reduced to a table of immediate measurement probabilities.

Z -I Q -0 Q

Input	Classical bit	Quantum output
0	0	$ 0\rangle$
1	1	$- 1\rangle$

6.4 Hadamard (H)



$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle.$$

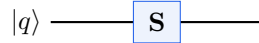
Hadamard changes between the computational z basis and the plus/minus x basis. It creates equal magnitudes from a basis state, but the sign of the second amplitude depends on the input. It is also self-inverse: applying H twice returns the original state. H is commonly used before controlled gates to create entanglement and after phase gates to reveal interference.

```
SET Q 0p
H -I Q -0 Q
MEASURE -I Q
```

Input	Measurement distribution	State before measurement
0	50% 0, 50% 1	$(0\rangle + 1\rangle)/\sqrt{2}$
1	50% 0, 50% 1	$(0\rangle - 1\rangle)/\sqrt{2}$

Equal classical probabilities hide the relative minus sign responsible for interference.

6.5 S phase (S)



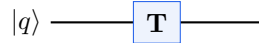
$$S = P(\pi/2) = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad S|1\rangle = i|1\rangle, \quad S^2 = Z.$$

S is a quarter-turn around the Bloch sphere's z-axis. It changes $|+\rangle$ into $(|0\rangle + i|1\rangle)/\sqrt{2}$ while leaving basis measurement probabilities unchanged. Its inverse is $S^\dagger = P(-\pi/2)$, not S itself.

```
S -I Q -0 Q
```

Input	Measured bit	Quantum output
0	0	$ 0\rangle$
1	1	$i 1\rangle$

6.6 T phase (T)



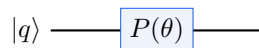
$$T = P(\pi/4) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}, \quad T^2 = S, \quad T^4 = Z.$$

T is an eighth-turn around z. Together with gates such as H and CNOT, it is an important non-Clifford resource for expressing general quantum computations. Its inverse is $T^\dagger = P(-\pi/4)$. `Tdg` and `Tinv` lower to that gate, and `Sdg` and `Sinv` lower to $S^\dagger = P(-\pi/2)$. `PHASEdg=pi/4` is the same negative phase.

```
T -I Q -0 Q
```

Input	Measured bit	Quantum output
0	0	$ 0\rangle$
1	1	$e^{i\pi/4} 1\rangle$

6.7 Arbitrary phase (PHASE)



$$P(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}, \quad \alpha|0\rangle + \beta|1\rangle \mapsto \alpha|0\rangle + \beta e^{i\theta}|1\rangle.$$

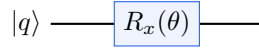
PHASE is the general form behind S, T, and Z. It preserves $|\alpha|^2$ and $|\beta|^2$ because $|e^{i\theta}| = 1$, yet rotates the relative phase by θ . On the Bloch sphere this is a z-axis rotation up to global phase. A later H or another basis-changing gate is normally needed to turn that phase difference into changed computational-basis probabilities. The angle is embedded in the opcode:

```
PHASE=1.5708 -I Q -0 Q      # radians
PHASE=3/2 -I Q -0 Q        # rational radians
PHASE=90d -I Q -0 Q        # degrees
PHASE=45/2d -I Q -0 Q      # rational degrees
PHASE=pi -I Q -0 Q
PHASE=-11pi/6 -I Q -0 Q
PHASE=2*pi/3 -I Q -0 Q
PHASEdg=pi/4 -I Q -0 Q     # P(-pi/4)
invPHASE=pi/2 -I Q -0 Q    # P(-pi/2)
```

Accepted numeric atoms include signed integers, decimals, leading-dot decimals, and one optional rational denominator. Pi notation accepts an optional sign, an integer coefficient, an optional *, and an optional positive integer denominator. A zero denominator and malformed angle raise an error.

Input	Measured bit	Quantum output
0	0	$ 0\rangle$
1	1	$e^{i\theta} 1\rangle$

6.8 X rotation (RX)

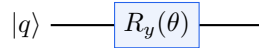


$$R_x(\theta) = \begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}.$$

RX rotates about the Bloch x -axis. `RX=pi` is X up to global phase.

```
RX=pi/2 -I Q -0 Q
RX=90d -I Q -0 Q
RXdg=pi/4 -I Q -0 Q
```

6.9 Y rotation (RY)

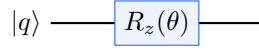


$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}.$$

RY rotates about y and is useful for preparing real superpositions from $|0\rangle$.

```
RY=pi/2 -I Q -0 Q
```

6.10 Z rotation (RZ)



$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}.$$

RZ rotates about z . It differs from PHASE by a global phase that does not affect single-qubit measurement probabilities.

```
RZ=pi/4 -I Q -0 Q
```

6.11 Dagger and inverse syntax

A reversible gate accepts **dg** (dagger) or **inv** (inverse), written after the opcode or before it. Both spellings set **reverse=true**. Self-inverse gates such as X, CNOT, AND, and OR still accept the marker so intent is visible, even though the matrix is unchanged.

```
Xdg -I Q -0 Q
invH -I Q -0 Q
CNOTdg -I A -0 T
ANDdg -I A B -0 T
dgSWAP -I A B -0 A B
Sdg -I Q -0 Q           # P(-pi/2)
Tinv -I Q -0 Q          # P(-pi/4)
PHASEdg=pi/4 -I Q -0 Q  # P(-pi/4)
```

Sdg, Sinv, dgS, and invS synthesize $S^\dagger$. Tdg, Tinv, dgT, and invT synthesize $T^\dagger$. PHASEdg=angle, PHASEinv=angle, dgPHASE=angle, and invPHASE=angle negate the angle. For these self-inverse gates the marker emits the ordinary gate, which is already its own inverse:

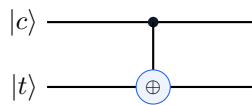
$X, Y, Z, H, \text{CNOT}, \text{CCNOT}, \text{CZ}, \text{CY}, \text{SWAP}.$

On the canvas the dagger prints $\dagger$ and uses a blue outline. While that step is active the outline is purple. A forward gate uses a red outline only while its step is active. The palette **Inverse** toggle drops the dagger instead of the forward gate.

A **dg** or **inv** marker on a reversible derived gate such as ANDdg is accepted and is self-inverse. MEASUREdg parses and warns, because measurement has no inverse. Compiler-internal RESET remains unavailable in source.

7 Controlled and two-qubit primitive gates

7.1 Controlled NOT (CNOT)



$$|c, t\rangle \mapsto |c, t \oplus c\rangle.$$

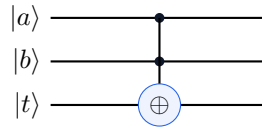
CNOT leaves the control unchanged. When the control is zero, the target is unchanged; when the control is one, X acts on the target. Because the map is a permutation of basis states, it is reversible and self-inverse. A superposition control can entangle the wires, so “classical XOR” describes only its basis-state action.

```
CNOT -I Control -0 Target
```

c	t	c'	t'
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

CNOT is a reversible embedding of XOR because it preserves the control. In the workbench, the control comes from **Control A** and the target from **Target particle**; **Control B** is ignored.

7.2 Doubly controlled NOT / Toffoli (CCNOT)



$$|a, b, t\rangle \mapsto |a, b, t \oplus (a \wedge b)\rangle.$$

CCNOT, also called Toffoli, applies X only when both controls are one. It is a reversible universal building block for classical computation embedded in a quantum circuit. The target's previous value matters: zero computes AND, while one computes NAND. The gate is self-inverse.

CCNOT -I A B -0 Target

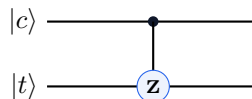
a	b	t	a'	b'	t'
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

With a target initialized to zero, Toffoli computes classical AND reversibly. This is the *equal inputs and outputs* requirement for a reversible gate in action: the two-input classical AND has no inverse, but its three-wire Toffoli embedding does, because no wire is discarded (*Requirements for a reversible gate* in the Theory Guide's mathematical foundations chapter).

Controls are preserved, the target is the output

In the table, the a' and b' columns always equal a and b : the two controls are only read. Only t can change, and it flips exactly in the two rows where $a = b = 1$. In the **Interactive workbench**, choose **CCNOT** in **Gate**, the two inputs in **Control A** and **Control B**, and the output wire in **Target particle**. The *CCNOT demo* starter circuit is exactly this gate with both controls prepared as 1.

7.3 Controlled Z (CZ)



$$\text{CZ}|c, t\rangle = (-1)^{ct}|c, t\rangle, \quad \text{CZ} = \text{diag}(1, 1, 1, -1).$$

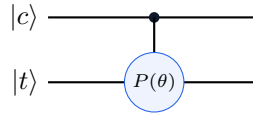
CZ adds a minus sign only to the $|11\rangle$ amplitude. No computational-basis bit changes, but superpositions can acquire observable relative phase. CZ is symmetric between its two wires, even though QPU syntax writes one as the input control and one as the output target. It is self-inverse.

CZ -I Control -0 Target

Input	Measured output	Amplitude multiplier	State
00	00	+1	$ 00\rangle$
01	01	+1	$ 01\rangle$
10	10	+1	$ 10\rangle$
11	11	-1	$- 11\rangle$

The classical columns are unchanged; the quantum phase is the operation.

7.4 Controlled phase (CPHASE)

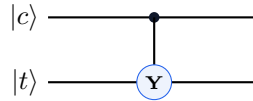


$$|11\rangle \mapsto e^{i\theta}|11\rangle,$$

while $|00\rangle$, $|01\rangle$, and $|10\rangle$ are unchanged. CZ is the special case $\theta = \pi$. CPHASE unlocks compact Quantum Fourier Transform and phase-estimation demos.

CPHASE=pi/2 -I Control -0 Target
CPHASE=90d -I Control -0 Target

7.5 Controlled Y (CY)



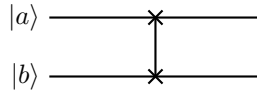
$$|0, t\rangle \mapsto |0, t\rangle, \quad |1, 0\rangle \mapsto i|1, 1\rangle, \quad |1, 1\rangle \mapsto -i|1, 0\rangle.$$

CY applies the Pauli Y rotation to the target only when the control is one. Its basis-bit pattern resembles CNOT, but the $\pm i$ factors distinguish their behavior on superpositions. Since $Y^2 = I$, CY is self-inverse.

CY -I Control -0 Target

c	t	c'	t'	Quantum phase
0	0	0	0	+1
0	1	0	1	+1
1	0	1	1	i
1	1	1	0	$-i$

7.6 Swap (SWAP)



$$\text{SWAP}|a, b\rangle = |b, a\rangle.$$

SWAP exchanges the complete quantum states of two wires, including amplitudes and any correlations with the rest of the register. It does not copy either state. Three CNOTs can decompose SWAP, but the QPU system provides it as one primitive instruction. Applying SWAP twice restores the original ordering.

```
SWAP -I A B -O A B
```

The two output tokens are mandatory and must resolve to the same two wires, in the same order, as the inputs.

a	b	a'	b'
0	0	0	0
0	1	1	0
1	0	0	1
1	1	1	1

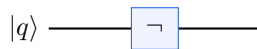
8 Derived logical gates

Derived gates are QPU instructions implemented as reversible target mutations. Inputs are preserved. If target t starts in zero, AND/OR/XOR write the familiar classical function. If it does not, the true behavior is the XOR embedding shown below.

One shape for every two-input gate

AND, NAND, OR, and XOR all follow $t' = t \oplus f(a, b)$. The two inputs a and b (**Control A** and **Control B** in the workbench) are read and left unchanged. The target t (**Target particle**) is the output workspace and is the only wire that changes. Only f differs between the four gates. To get the ordinary classical answer, start the target at 0 with SET Out 0p.

8.1 Logical NOT (NOT)



$$t' = t \oplus 1.$$

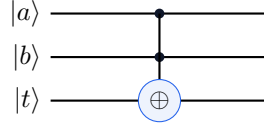
NOT is the derived logical spelling of a bit flip. In the current simulator it uses the same X matrix on the output target. The required input establishes the one-input instruction shape, but a distinct input is not copied into the output; use the same token on both sides.

```
NOT -I Target -O Target
```

t	t'
0	1
1	0

The current gate implementation mutates the output target. Use the same token for input and output; a distinct input token is not copied.

8.2 Reversible AND (AND)



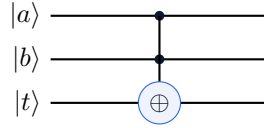
$$t' = t \oplus (a \wedge b).$$

AND flips the target exactly on the basis components where both controls are one. This is Toffoli-style behavior. It remains linear on superpositions and may entangle the target with its controls. Calling it AND is convenient only when the target is deliberately initialized.

```
SET Out Op
AND -I A B -O Out
```

a	b	$t = 0$	$t' = a \wedge b$
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	1

8.3 Reversible NAND (NAND)



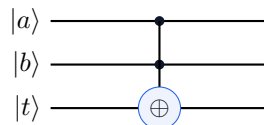
$$t' = t \oplus (a \wedge b) \oplus 1.$$

NAND first performs the all-controls-active mutation and then inverts the target. With target zero it produces the familiar NAND column. With target one it instead produces AND, illustrating why reversible target state must always be included in the full mathematical definition.

```
SET Out Op
NAND -I A B -O Out
```

a	b	$t = 0$	$t' = \neg(a \wedge b)$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	0	0

8.4 Reversible OR (OR)



$$t' = t \oplus (a \vee b).$$

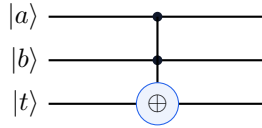
OR uses an “any control is one” predicate. If either or both controls are active, the target is flipped. This direct derived operation is equivalent on basis states to combining XOR and AND through $a \vee b = a \oplus b \oplus (a \wedge b)$.

```
SET Out Op
OR -I A B -O Out
```

a	b	$t = 0$	$t' = a \vee b$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	0	1

For bits, $a \vee b = a \oplus b \oplus (a \wedge b)$.

8.5 Reversible XOR (XOR)



$$t' = t \oplus a \oplus b.$$

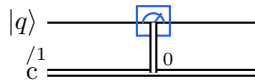
XOR flips the target when an odd number of controls is one. With two inputs, this is the ordinary exclusive-or predicate. The parity definition explains why the 11 row does not flip: two active controls have even parity.

```
SET Out Op
XOR -I A B -O Out
```

a	b	$t = 0$	$t' = a \oplus b$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	0	0

9 Measurement gate

9.1 Measurement (MEASURE)



```
MEASURE -I Q           # exactly one named wire
MEASURE                # every wire known at this point
MEASURE -I Q -BASIS X  # measure the X observable instead of Z
```

MEASURE takes no -O. Repeated one-wire commands are useful when only declared returns should become classical. A bare command emits one measurement operation per currently allocated wire, and a bare MEASURE -BASIS Y measures every one of them in the Y basis.

Without **-BASIS** the wire is read in the computational (Z) basis, as below. **-BASIS X** and **-BASIS Y** read another observable: the recorded bit 0 means $|+\rangle$ (or $|+i\rangle$), 1 means $|-\rangle$ (or $|-i\rangle$), and the wire is left in that state. **-BASIS Z** is the default written out. The canvas marks an X or Y meter with a small X or Y.

State	Outcome probabilities	Collapsed state
$ 0\rangle$	$P(0) = 1$	$ 0\rangle$
$ 1\rangle$	$P(1) = 1$	$ 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$P(0) = \alpha ^2, P(1) = \beta ^2$	$ 0\rangle$ or $ 1\rangle$

Three ways to measure in the builder

The meter A step *in the circuit*, drawn on the qubit wire. A double stroke drops from that column to the **c** lane and numbers the bit that was read. **c** records the time of the measurement. The qubit wire stays a single line, and later gates on that particle still run. The meter is saved in the protocol as **MEASURE**.

Measure target Measures the selected **Target particle** once, right now. Nothing is added to the circuit.

Measure all Measures every wire not yet measured, right now.

MEASURE, like RESET, is not reversible: once a superposition collapses, the lost amplitudes cannot be recovered. Choose **Reset state** to rewind and sample again. This is the *isolation from measurement* requirement for a reversible gate: MEASURE is not linear and has no inverse matrix, so it sits outside the closed-system unitary evolution the other three requirements assume (*Requirements for a reversible gate* in the Theory Guide’s mathematical foundations chapter).

10 Reset: an operation that is not a gate

The word *reset* appears in two unrelated places: the compiler’s internal **RESET** operation, and three buttons in the Circuit builder. Neither is an ordinary quantum gate.

10.1 Reversible gates versus RESET

Every gate in the palette except the meter is *reversible*: its effect can always be undone. X, Y, Z, H, CNOT, CCNOT, CZ, CY, SWAP, NOT, AND, NAND, OR, and XOR undo themselves when applied a second time. S, T, and PHASE are undone by the same phase with the opposite sign. Mathematically each is a *unitary* matrix U , and $U^\dagger$ undoes it. **RESET** is different. It forces one wire to $|0\rangle$ whatever it held before:

$$|0\rangle \mapsto |0\rangle, \quad |1\rangle \mapsto |0\rangle, \quad \alpha|0\rangle + \beta|1\rangle \mapsto |0\rangle.$$

Two different inputs, $|0\rangle$ and $|1\rangle$, give the same output, so nothing can tell afterwards which one was there. The information has been erased, and no matrix U can undo that. **RESET** is therefore not unitary, just like **MEASURE**. *Requirements for a reversible gate*, in the Theory Guide’s mathematical foundations chapter, states all four conditions a gate must meet to be reversible; **RESET** and **MEASURE** are the only two operations in QPU that fail them.

	Reversible gates (X, H, MEASURE, CNOT, AND, ...)		RESET
Unitary?	Yes	No	No
Can be undone?	Yes, by the inverse gate	No	No
Result depends on chance?	No	Yes	Only for the other wires of an entangled wire
Written in source?	Yes	Yes	No — write SET wire 0p
Shown on canvas?	Yes. Inverse copies add †, a blue outline, and purple when active.	Yes, as the meter on the qubit, with a double stroke down to c	No (hidden)

10.2 Where RESET comes from

You never type RESET. The compiler inserts it when a wire must start at zero:

- `SET Out 0p` on a workspace wire. (`SET Out 1p` and `SET Out sp` are instead prepared by applying X or H to a wire that starts at 0.)
- `INCREASECYCLE`, which zeroes pending workspace before the next stage.
- `RUNCHILD` and `CALL`, which zero each parent output wire once before expanding the child.

RESET steps are hidden from the canvas and filtered from the **Execution log** so that the visible circuit shows only the gates you wrote. They still run during simulation.

Reset a wire only when it is not entangled

RESET behaves like real hardware: it measures the wire in the Z basis and, if the result is 1, flips it back with X. The reset wire always ends at $|0\rangle$. If the wire was already definitely 0 or 1, nothing random happens. If it was in a superposition of its own, such as $|+\rangle$, it still ends at $|0\rangle$ and the other wires are unaffected. But if it was entangled with other wires, the hidden measurement collapses them too. For example, resetting one half of a Bell pair leaves the other half as 0 or 1 at random, each half the time, and no longer entangled. Averaged over many runs this is exactly the physical reset described in the Theory Guide's chapter *Mixed states, entanglement measures, and noise*. In multi-stage circuits, *uncompute* temporary wires back to 0 with reversible gates instead of relying on a reset.

10.3 Reset buttons compared

None of these buttons inserts a RESET operation into your circuit. They only change what the page is showing and simulating.

Kept or cleared?	Reset state	Clear circuit	Reset site	Notes
Gates on the canvas	kept	cleared	cleared	
Number of wires	kept	kept	back to 3	
Start states (<i>name</i> start)	kept	kept	all 0p	
Protocol editor text	kept	rewritten	default	Clear circuit writes an empty canvas protocol into the editor.
Compiled PARAMS / RETURNVALS	kept	cleared	cleared	
Quantum state and measurements	rewound	rewound	rewound	Back to the start states; log restarts.
Workbench selections	kept	kept	defaults	Gate H, target q0, controls q1 and q2, phase 90°.
Custom gates and catalog	kept	kept	kept	Nothing removes these except Remove or closing the browser session.

Which one do I want?

Reset state “Run the same experiment again.” Use it between repeated runs of a circuit that measures, to collect several samples.

Clear circuit “Keep my wires, start a new circuit.”

Reset site “Put everything back the way it was when I first opened the page.” The **Reset site** entry in the navigation menu and **Reset site completely** on the **More** page do the same thing.

11 Mixed states, entanglement measures, and noise

Every simulation in QPU passes through one *physics engine*. The compiler turns source into gates and the simulator decides the order in which they run, but the physics engine alone changes the quantum state and computes the quantities in this chapter. The particle cards in the Circuit builder show several of them.

11.1 Density matrices

A pure state $|\psi\rangle$ can also be written as the *density matrix*

$$\rho = |\psi\rangle\langle\psi|, \quad \rho_{ij} = a_i \overline{a_j}.$$

The diagonal entries are the basis probabilities. The off-diagonal entries, called *coherences*, carry the relative phases that make interference possible. A *mixed state* is a weighted mixture of pure states,

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \quad p_k \geq 0, \quad \sum_k p_k = 1,$$

and describes either classical uncertainty about which state was prepared or a part of a larger entangled system. Every density matrix is Hermitian, has trace 1, and has no negative eigenvalues. Global phase cancels in $|\psi\rangle\langle\psi|$, so a density matrix records exactly the physically observable part of a state.

11.2 Reduced states and the partial trace

A single wire of an entangled register has no state vector of its own, but it always has a density matrix. For a system AB the state of A alone is the *partial trace*

$$\rho_A = \text{Tr}_B(\rho_{AB}), \quad (\rho_A)_{ij} = \sum_k (\rho_{AB})_{(i,k),(j,k)}.$$

It gives the correct statistics for every measurement made on A alone. For one qubit it fixes a *Bloch vector* (x, y, z) through

$$\rho = \frac{1}{2} (I + xX + yY + zZ), \quad x = \langle X \rangle, \quad y = \langle Y \rangle, \quad z = \langle Z \rangle.$$

Pure states lie on the Bloch sphere ($r = 1$); mixed states lie inside the Bloch ball ($r < 1$). The particle cards draw each wire from its reduced state, which is why an entangled wire's arrow shrinks toward the centre.

11.3 Purity and mixedness

The purity of a state is

$$P = \text{Tr}(\rho^2) = \sum_{ij} |\rho_{ij}|^2.$$

$P = 1$ exactly for pure states. For one qubit $P = (1 + r^2)/2$, which ranges from 1 on the sphere to $\frac{1}{2}$ at the centre (the maximally mixed state $I/2$). The particle card's *mixedness* is the normalized linear entropy $M = 2(1 - P)$: 0 for a pure wire and 1 for a maximally mixed one.

Mixed does not mean noisy

Without noise, the whole register is always in a pure state. A wire can still look mixed because it is entangled with other wires: its information lives in the correlations, not in the wire alone. When the register is pure and a wire is mixed, the card shows **Entangled with register**.

11.4 Entropy and entanglement

The *von Neumann entropy* measures how mixed a state is, in bits:

$$S(\rho) = -\text{Tr}(\rho \log_2 \rho) = -\sum_k \lambda_k \log_2 \lambda_k,$$

where λ_k are the eigenvalues of ρ . A pure state has $S = 0$; a maximally mixed qubit has $S = 1$.

If the whole register is pure, a subsystem A is entangled with the rest exactly when ρ_A is mixed, and $S(\rho_A)$ is the *entanglement entropy*: how many bits of entanglement join A to the rest.

State	One qubit	Two qubits together	Whole register
$ 00\rangle$	$P = 1, S = 0$	$P = 1$	pure
Bell $(00\rangle + 11\rangle)/\sqrt{2}$	$P = \frac{1}{2}, S = 1$	$P = 1$	pure
GHZ $(000\rangle + 111\rangle)/\sqrt{2}$	$P = \frac{1}{2}, S = 1$	$P = \frac{1}{2}, S = 1$	pure

The GHZ row shows why entanglement cannot be reduced to pairwise links. Any two GHZ qubits are only classically correlated (00 or 11 with equal chance), yet all three together form a single pure, entangled state. Questions about entanglement are therefore always about a chosen split of the register into a subsystem and the rest.

When the whole register is mixed, for example after noise, local mixedness no longer proves entanglement. The physics engine then uses the *partial transpose* test: transpose only B 's indices of ρ_{AB} . If the result has a negative eigenvalue the state is entangled, and the sum of the negative eigenvalues' magnitudes is the *negativity* ($\frac{1}{2}$ for a Bell pair). A negative eigenvalue is proof for any split. The converse holds only for two qubits (or a qubit and a qutrit): for larger splits some entangled states have no negative eigenvalue at all. The engine therefore reports one of three results, and says which test it used:

Result	Meaning
entangled	Proven: the reduced state of a pure register is mixed, or the partial transpose has a negative eigenvalue.
separable	Proven: the reduced state of a pure register is pure, or a two-qubit mixed state has no negative eigenvalue.
inconclusive	Not decided: a larger mixed split passed the partial-transpose test, or the register is too large to test. This is <i>not</i> evidence that the wires are unentangled.

A particle card shows **Entangled with register** only for a proven result and **Entanglement undetermined** for an inconclusive one; it never labels a wire “not entangled” because a test failed to find entanglement.

QPU has no ENTANGLE instruction. Entanglement always comes from ordinary gates such as H followed by CNOT; the physics engine only detects and measures it.

11.5 Measurement in other bases

MEASURE measures in the computational (Z) basis. The physics engine can also measure the X and Y observables by rotating the wire's eigenbasis onto Z, measuring, and rotating back:

Basis	Outcome 0	Outcome 1	Rotation onto Z
Z	$ 0\rangle$	$ 1\rangle$	none
X	$ +\rangle$	$ -\rangle$	H
Y	$ +i\rangle$	$ -i\rangle$	$HS^\dagger$

The state $|+\rangle$ gives a random result in the Z basis but always gives 0 in the X basis: what a measurement reveals depends on the observable. In QPU source, `MEASURE -I Q -BASIS X` (or Y) chooses the observable; a `MEASURE` without `-BASIS` is a Z measurement.

11.6 Fidelity

Fidelity measures how close a simulated state ρ is to an intended state σ :

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2, \quad F(|\psi\rangle, |\phi\rangle) = |\langle \psi | \phi \rangle|^2.$$

$F = 1$ means the states are identical (up to global phase) and $F = 0$ means they are perfectly distinguishable. For example, $F(|+\rangle, |0\rangle) = \frac{1}{2}$.

11.7 Reset as a physical operation

RESET forces a wire to $|0\rangle$. As a density-matrix operation it is the *reset channel*: the wire ends in $|0\rangle$ and whatever it shared with the other wires is traced out,

$$\rho \mapsto |0\rangle\langle 0| \otimes \text{Tr}_q(\rho).$$

A state vector cannot hold the mixed result of resetting an entangled wire, so a state-vector simulation follows one run of this channel the way real hardware does: measure the wire in Z, then apply X if the result was 1. Averaged over many runs this gives exactly the reset channel. The Reset chapter describes what this means for circuits.

11.8 Noise and decoherence

Real qubits are not isolated. Their interaction with the environment is described by *quantum channels*

$$\rho \mapsto \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I,$$

which turn pure states into mixed ones. The physics engine provides these single-qubit channels:

Channel	Effect
Bit flip (p)	$\rho \mapsto (1-p)\rho + pX\rho X$: $ 0\rangle$ and $ 1\rangle$ swap with probability p .
Phase flip (p)	$\rho \mapsto (1-p)\rho + pZ\rho Z$: coherences shrink by $1-2p$.
Depolarizing (p)	$\rho \mapsto (1-p)\rho + pI/2$: the Bloch vector shrinks by $1-p$.
Amplitude damping (γ)	Energy loss: $ 1\rangle$ decays to $ 0\rangle$ with probability γ .
Phase damping (λ)	Coherences shrink by $\sqrt{1-\lambda}$; populations are unchanged.

Decoherence is usually quoted as two times. T_1 is the relaxation time: after time t an excited qubit remains in $|1\rangle$ with probability e^{-t/T_1} . T_2 is the coherence time: off-diagonal entries decay as e^{-t/T_2} . Physics requires $T_2 \leq 2T_1$.

Cycles are not time

QPU cycles (`INCREASECYCLE`) are logical program stages, not elapsed physical time. Decoherence uses a separate timing model that assigns a duration to each gate. Noise is always optional: ordinary runs are ideal and use the cheaper state vector (2^n amplitudes) rather than a density matrix (4^n entries). Choose **Run with noise** in the **Physics inspector** on the Particle visualization page to rerun a circuit with a noise channel and optional T_1/T_2 ; the inspector reports the fidelity with the ideal run. Noise settings are not part of QPU source.

11.9 No cloning and teleportation

CNOT copies a known basis value onto a $|0\rangle$ target, but applied to $|+\rangle|0\rangle$ it produces a Bell pair, not $|+\rangle|+\rangle$. The two results have fidelity $\frac{1}{2}$ and the target becomes entangled rather than a copy. This is the *no-cloning theorem*: no operation can copy an unknown quantum state.

Quantum state can still be *moved*. SWAP exchanges two wires' states, and teleportation transfers a state using only ordinary operations: share a Bell pair (H, CNOT), entangle the message with one half (CNOT, H), measure two wires, and apply X and Z to the destination under classical **-IF** conditions. The destination then holds the original state with fidelity 1 and the source wire has been measured, so the state was never in two places at once. QPU has no **CLONE** or **TELEPORT** instruction.

12 Energy, frequency, and control pulses

Gates in QPU are normally exact matrices. Real qubits are physical two-level systems with an energy gap, and gates are made by driving them with microwave pulses. The physics engine can model this level of detail. Frequency is never a free display value: it is derived from each qubit's energy levels and it drives the evolution.

Frequencies below are in GHz and times in ns, so $2\pi ft$ is a plain angle. Energies use SI units.

12.1 Energy levels and the Planck–Einstein relation

A photon of frequency f carries energy $E = hf = \hbar\omega$ with $\omega = 2\pi f$. A qubit absorbs or emits such a photon when it moves between its levels, so its *transition frequency* is

$$f_{01} = \frac{E_1 - E_0}{h}.$$

The engine gives each modelled wire the idle Hamiltonian

$$H_0 = -\frac{\hbar\omega_{01}}{2}Z,$$

whose levels are $E_0 = -\hbar\omega_{01}/2$ and $E_1 = +\hbar\omega_{01}/2$, a gap of exactly $\hbar\omega_{01}$. A typical superconducting qubit at $f_{01} = 5$ GHz has $\Delta E \approx 3.3 \times 10^{-24}$ J (about 21 μeV), and the matching photon has a wavelength of about 6 cm.

12.2 Free precession and the physical clock

A qubit left alone still evolves: $e^{-iH_0t/\hbar}$ gives $|1\rangle$ a phase relative to $|0\rangle$ that changes as

$$\Delta\phi = -\omega_{01} \Delta t,$$

so the Bloch vector precesses about the z-axis at f_{01} . Populations do not change, which is why idle precession is invisible to Z measurements but matters for everything that depends on relative phase.

Physical simulation keeps a *physical clock*. Every operation lasts its physical duration, and during that time every wire evolves under its own Hamiltonian while T_1 and T_2 act over the same Δt . Logical cycles (`INCREASECYCLE`) never advance this clock.

12.3 Frames

At several GHz the lab-frame phase turns billions of times a second. Most calculations therefore use a *rotating frame* that turns at each qubit's nominal f_{01} . In that frame a perfectly calibrated idle qubit does nothing, and only errors remain. A `frequencyOffset` (calibration error) or a `frequencyDriftRate` (slow drift) leaves a residual precession

$$\Delta\phi = -2\pi \int (f_{\text{actual}}(t) - f_{\text{nominal}}) dt,$$

which becomes phase error in the circuit. The lab frame shows the full precession.

12.4 Driving a qubit: resonance, Rabi, and detuning

A microwave drive of carrier frequency ω_d adds

$$H_d(t) = \hbar\Omega(t) \cos(\omega_d t - \varphi) X,$$

where Ω is the Rabi frequency, set by the drive amplitude, and φ is the drive phase. In the frame rotating with the drive, the fast terms average out and

$$H \approx \frac{\hbar}{2} (\Delta Z + \Omega(\cos \varphi X + \sin \varphi Y)), \quad \Delta = \omega_d - \omega_{01}.$$

- **Resonance** ($\Delta = 0$). The qubit rotates about an axis in the XY plane at the Rabi frequency, and the rotation angle is $\theta = \Omega t$ for a square pulse. A π -pulse ($\Omega t = \pi$, $\varphi = 0$) is an X gate up to global phase, and $\varphi = \pi/2$ gives Y. Starting from $|0\rangle$, $P(1) = \sin^2(\Omega t/2)$: these are *Rabi oscillations*.
- **Detuning** ($\Delta \neq 0$). The axis tilts toward z and the rotation speeds up to the *generalized Rabi frequency* $\Omega' = \sqrt{\Omega^2 + \Delta^2}$. Starting from $|0\rangle$,

$$P(1) = \frac{\Omega^2}{\Omega^2 + \Delta^2} \sin^2 \frac{\Omega' t}{2},$$

so a detuned drive never completes the flip.

- **Far off resonance** ($|\Delta| \gg \Omega$). The drive hardly moves population but shifts the qubit frequency by $\delta f \approx -\Omega^2/(2\Delta)$ (in the same frequency units): the *AC Stark shift*.

A calibrated pulse sets Ω so its area gives the requested angle against the *nominal* f_{01} . A qubit whose real frequency has drifted is then detuned from it, which is how calibration error becomes gate error.

12.5 Pulse shapes

Pulses have an envelope $\varepsilon(t)$ with peak 1. A *square* pulse is the simplest. A *Gaussian* pulse has a narrower spectrum, so it disturbs neighbouring frequencies less. A *DRAG* pulse adds a derivative quadrature component that suppresses leakage into a third level. The engine's qubits have only two levels, so DRAG there adds a small error instead of removing one. The rotation of a resonant pulse is Ω times the area under $\varepsilon(t)$.

12.6 The rotating-wave approximation

Dropping the fast terms above is the *rotating-wave approximation* (RWA). It is accurate when $\Omega \ll \omega_{01}$ and it lets the engine take steps set by Ω and Δ (MHz) instead of by the carrier (GHz). The engine applies it only when asked (approximation: 'rwa'); the 'exact' strategy integrates the full time-dependent Hamiltonian with small steps and agrees with the RWA to order Ω/ω_{01} .

12.7 Coupled qubits

Entanglement between physical qubits comes from interaction Hamiltonians, not from a named gate:

Coupling	Effect
$ZZ: \hbar 2\pi J (Z \otimes Z)/4$	A conditional phase of $2\pi J$ per unit time. Two $ +\rangle$ qubits become maximally entangled after $t = 1/(2J)$, which is a CZ gate up to single-qubit phases.
Exchange: $\hbar 2\pi J (X \otimes X + Y \otimes Y)/2$	Swaps $ 01\rangle$ and $ 10\rangle$. Full exchange takes $t = 1/(4J)$, and half of it entangles. Qubits whose frequencies differ by much more than J barely exchange.

An always-on coupling that nobody asked for acts during every idle slot. This *crosstalk* can entangle qubits without any two-qubit gate in the circuit.

12.8 Temperature

A qubit exchanges energy with a bath at temperature T . In equilibrium its excited population is the Boltzmann value

$$p_1 = \frac{1}{1 + e^{\hbar f_{01}/k_B T}},$$

tiny at 20 mK and 5 GHz but not zero. With a temperature set, T_1 relaxation pulls the qubit toward p_1 instead of toward $|0\rangle$ (generalized amplitude damping).

12.9 Wavelengths

A drive photon has wavelength $\lambda = c/f$. Massive particles have the de Broglie wavelength $\lambda = h/p$; an electron at 10^6 m/s has $\lambda \approx 0.73$ nm. The Physics inspector shows it for teaching, but it does not affect gate execution, because QPU simulates information-carrying states rather than particles moving through space.

What is not modelled

Qubits here have exactly two levels, so leakage to $|2\rangle$ and anharmonicity are not simulated. Noise is described by channels and T_1/T_2 , not by a frequency-dependent noise spectrum $S(\omega)$. Spatial wave packets are outside the gate-level simulator.

13 Glossary

Term	Meaning
Amplitude	A complex coefficient of a basis state. Its squared magnitude contributes to measurement probability.
Ancilla	An auxiliary qubit or workspace wire, usually prepared in a known state, used to hold intermediate information in a reversible computation.
Basis state	A state used as a coordinate direction. The computational basis for one qubit is $ 0\rangle, 1\rangle$.
Bloch sphere	A geometric representation of pure one-qubit states after ignoring global phase. North/south are $ 0\rangle/ 1\rangle$; longitude represents relative phase.
Boolean algebra	The mathematics of 0/1 values and operations such as NOT, AND, OR, and XOR.
Child process	A separately named protocol expanded into a parent through RUNCHILD or CALL.
Combinational logic	A classical circuit whose output depends only on current inputs, not stored history.
Compile-time recursion	Bounded self-expansion of a child process marked REC or TREC. RECUR expands another frame with DEPTH-1; EXIT WHEN ends a frame. Tail-position REC auto-converts to TCO (iterative one-frame expansion, F#-style); TREC requires that form. No runtime loop or recursive QPU execution is created.
Control	A wire whose basis condition decides whether a controlled operation acts on its target. The control itself is preserved by the standard controlled gates.
Dn badge	Teal label above a recursive call on the canvas showing the current DEPTH; counts down while stepping and hides when the call ends.
Decoherence	Loss of quantum coherence through contact with the environment, usually quoted as a relaxation time T_1 and a coherence time T_2 . It turns pure states into mixed ones.
Density matrix	The matrix ρ describing a pure or mixed state. For a pure state $\rho = \psi\rangle\langle\psi $; its diagonal holds basis probabilities.
DEPTH	Read-only remaining recursive expansions for the current child frame. Supplied by RUNCHILD -DEPTH and decremented by RECUR.
Detuning	$\Delta = \omega_d - \omega_{01}$, the difference between a drive and the qubit's transition frequency. A detuned drive rotates faster but never fully flips the qubit.
Entanglement	A joint multi-qubit state that cannot be written as a tensor product of independent qubit states.
Entanglement entropy	The von Neumann entropy of a subsystem of a pure register, in bits. 0 means unentangled; 1 for either half of a Bell pair.
Fan-out	Sending one classical value to multiple destinations. Unknown quantum states cannot be universally copied; CNOT copies only known basis information onto a zero target and otherwise may entangle.
Fidelity	How close two states are, from 0 (perfectly distinguishable) to 1 (identical up to global phase). For pure states $F = \langle\psi \phi\rangle ^2$.
Garbage	Reversible-workspace information no longer wanted as output but still present because it cannot simply be erased.
Gate	A state transformation. Closed-system quantum gates are unitary; measurement and reset have different semantics.
Global phase	A unit complex multiplier applied to the whole state. It does not change physical measurement predictions.
IF / ELSE / ENDIF	Structured classical feed-forward sugar. Compiles to flat gates carrying -IF conditions (ELSE gates get the opposite value) and branch metadata for the canvas labels. The header is either a measured bit (IF A=1) or a gate expression evaluated on a scratch copy, such as IF (AND -I A B -O B) = 1p. Both branches stay in the circuit; the one whose test fails is skipped and drawn faded as “skipped”. Never forks the quantum wire.
Ket	Dirac notation $ \psi\rangle$ for a state column vector.
Measurement	Conversion of quantum amplitudes into a sampled classical result, with state collapse and renormalization.

Term	Meaning
Mixed state	A state that is a probabilistic mixture of pure states, described by a density matrix rather than a state vector. A single wire of an entangled register is mixed even with no noise.
Noise channel	A non-unitary map $\rho \mapsto \sum_k K_k \rho K_k^\dagger$ modelling errors such as bit flips, phase flips, depolarization, or energy loss.
Phase	The angular part of a complex amplitude. Relative phase controls interference even when immediate basis probabilities are unchanged.
Physical clock	Elapsed physical time in a physical simulation. Gate durations advance it; logical cycles do not.
Process	A named QPU protocol body with ordered PARAMS and RETURNVALS.
Protocol	Plain-text QPU instructions describing preparation, gates, composition, measurement, and returned references.
Purity	$\text{Tr}(\rho^2)$: 1 for a pure state, down to $\frac{1}{2}$ for a maximally mixed qubit.
Qubit	A normalized two-component complex state $\alpha 0\rangle + \beta 1\rangle$, not merely an unknown classical bit.
Rabi frequency	Ω , how fast a resonant drive rotates a qubit; a π -pulse lasts π/Ω .
Reduced state	The density matrix of part of a register, obtained with the partial trace. It predicts every measurement made on that part alone.
Register	One or more logical wires considered together. Current JOIN/SPLIT syntax is accepted but does not yet lower to register operations.
Relative phase	The phase difference between amplitudes. Unlike global phase, it can change later interference and measurement probabilities.
Rotating frame	A description that turns with each qubit's nominal frequency, so a calibrated idle qubit stands still and only errors remain.
Rotating-wave approximation	Dropping the fast counter-rotating terms of a drive (RWA); accurate when the Rabi frequency is far below f_{01} .
Superposition	A coherent linear combination of basis states. sp prepares the equal plus superposition.
Target	The wire whose amplitudes a gate mutates; normally the first -0 reference.
Tensor product	The operation $\otimes$ that combines state spaces. Two one-qubit vectors produce a four-amplitude two-qubit vector.
Token	A source-level logical name resolved to a qubit wire or alias during compilation.
Transition frequency	fre- $f_{01} = (E_1 - E_0)/h$, the frequency of the photon a qubit absorbs or emits between its two levels.
Truth table	Listed input/output expectations for measured RETURNVALS. It describes basis behavior but not arbitrary phase or entanglement.
Uncomputation	Running suitable inverse gates in reverse order to return temporary workspace to a known state without irreversible erasure.
Unitary	A matrix satisfying $U^\dagger U = I$, preserving normalization and having a reversible inverse.